
Statistics 212: Lecture 12 (March 12, 2025)

Convergence to Brownian Motion in Path Space

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1 Ways to Get Extra Credit

- Tell Mark about typos in HW/notes
- Future extra credit problems

2 Donsker's Theorem

Theorem 2.1 (Donsker's Theorem). *Let $X_1, X_2, \dots$ be a simple random walk (i.e., $X_{i+1} - X_i = \pm 1$ iid) for $n \geq 1$. Consider the random function on $[0, 1]$, define on $\mathbb{Z}/n$ by*

$$W^{(n)}(k/n) = X_k / \sqrt{n}$$

and interpolated linearly in between. Then $W^{(n)} \xrightarrow{d} \text{Law}(BM)$ as $n \rightarrow \infty$. In other words, for any continuous $f : C([0, 1]) \rightarrow \mathbb{R}$, we see $\lim_{n \rightarrow \infty} E[f(W^{(n)})] = E[f(B)]$ where B is Brownian motion.

Remark. In fact, this theorem holds for IID sums of any mean 0 and variance 1 random variable. This implies the Central Limit Theorem for IID sums under the same assumptions. Namely fix $\phi : \mathbb{R} \rightarrow \mathbb{R}$ that's bounded + continuous. Now consider $f(W) = \phi(W(1))$ which is bounded and continuous from $C([0, 1]) \rightarrow \mathbb{R}$. This implies $E[\phi(W^{(n)}(1))] \rightarrow E[\phi(B(1))]$ which is a CLT as ϕ is arbitrary.

Proof. This theorem can be proven using the Wald identities. The idea is to use an explicit coupling between Brownian motion and a simple random walk. We first start with a Brownian motion $B_t \sim BM$ and construct a simple random walk out of it. Consider a sequence of stopping times $\tau_1 < \tau_2 < \dots$ where each stopping time corresponds to hitting an integer. More formally, τ_{i+1} is the first time $t \geq \tau_i$ with $|B_t - B_{\tau_i}| = 1$. (For the more general statement, one represents any mean 0 variance 1 random variable as a stopped Brownian motion, which is called the *Skorokhod embedding*.)

We claim that $\tau_1, \tau_2 - \tau_1, \tau_3 - \tau_2, \dots$ are iid each with mean 1. By the Strong Markov property, these are iid. Recall that $B_t^2 - t$ is a martingale, so by a Wald identity, we have $1 = E[B_{\tau_j}^2] = E[\tau_j]$. Thus, by the strong law of large numbers, we can immediately deduce that $\lim_{n \rightarrow \infty} \frac{\tau_n}{n} = 1$.

We also see that $B_{\tau_1}, B_{\tau_2}, \dots$ is a simple random walk. Let $X_j = B_{\tau_j}$. Define $W^{(n)}(k/n) = X_k / \sqrt{n}$ and $B^{(n)}(t/n) = B_t / \sqrt{n}$. We claim $d_{\text{sup}}(W^{(n)}, B^{(n)}) \xrightarrow{p} 0$ (which also implies convergence in distribution). To

prove this, we define a “re-parameterized BM” $\tilde{B}$ where

$$\tilde{B}^{(n)}(k/n) = B(\tau_k)/\sqrt{n} = B^{(n)}(\tau_k/n) = W^{(n)}(k/n)$$

where the path is linear in between defined points.

We claim $d_{\sup}(B^{(n)}, \tilde{B}^{(n)}) \xrightarrow{P} 0$. A bit more detail, we see for all $\epsilon > 0$, there exists a random δ where $\sup_{|s-t| \leq \delta} |B^{(n)}(t) - B^{(n)}(s)| \leq \epsilon$. We can choose a deterministic δ_* such that

$$P(\sup_{|s-t| \leq \delta_*} |B^{(n)}(t) - B^{(n)}(s)| \leq \epsilon) \geq 1 - \epsilon.$$

(Slight side tangent to expound on the above: For each ϵ, δ , let $A_{\epsilon, \delta} \subseteq C([0, 1])$ consist of functions with $\sup_{|s-t| \leq \delta} |B_t - B_s| \leq \epsilon$. Almost surely, we have Brownian motion is continuous, so $\forall \epsilon$, we take the largest δ so $B \in A_{\epsilon, \delta}$. We see $\forall \epsilon, P[\bigcup_{\delta=1/n} A_{\epsilon, \delta}] = 1$ for Brownian Motion, which implies there exists a δ_* with $P[\bigcup_{\delta \geq \delta_*} A_{\epsilon, \delta}] \geq 1 - \epsilon$.)

For this (ϵ, δ_*) , if n is large enough, then $P[\sup_{0 \leq k \leq n} \frac{|\tau_k - k|}{m} \leq \delta_*] \geq 1 - \epsilon$ by the law of large numbers. This implies almost sure convergence, which also implies convergence in probability. As a result, we can conclude that $P[d_{\sup}(B^{(n)}, \tilde{B}^{(n)}) \leq \epsilon + 2/\sqrt{n}] \geq 1 - 2\epsilon$. We have two sources of error, which is why we have $1 - 2\epsilon$, and $2/\sqrt{n}$ relaxes the bound for the times in between k/n . Thus, we see $d_{\sup}(B^{(n)}, \tilde{B}^{(n)}) \xrightarrow{P} 0$.

Furthermore, we can pretty easily see that $d_{\sup}(\tilde{B}^{(n)}, W^{(n)}) \leq 2/\sqrt{n}$. Combining these two facts together, we have

$$P(d_{\sup}(B^{(n)}, W^{(n)}) \geq \epsilon) \leq \epsilon$$

so $W^{(n)} \xrightarrow{P} BM$. Whew!

It is intuitive that this convergence in probability implies convergence in distribution. Let's work through it. Fix a $f : C([0, 1]) \rightarrow \mathbb{R}$ with continuity and boundedness. From the definition of continuity, there exists δ such that $|f(B) - f(W)| \leq \epsilon$ if $d_{\sup}(B, W) \leq \delta$. Similarly to before, given f , we can choose a deterministic δ_* so

$$P(\sup_{W; d_{\sup}(W, B) \leq \delta} |f(B) - f(W)| \leq \epsilon) \geq 1 - \epsilon.$$

So for large n with probability $1 - \delta_*$, we have $d_{\sup}(B^{(n)}, W^{(n)}) = \delta_*$ which implies with probability $1 - \delta_* - \epsilon$, we have $\sup_W |f(B^{(n)}) - f(W)| \leq \epsilon$ and $|f(B^{(n)}) - f(W^{(n)})| \leq d_{\sup}(W, f) \leq d_*$. This further implies that $E[f(B^{(n)})] \rightarrow E[f(W^{(n)})]$. $\square$

This theorem introduces more questions:

- How do we think about convergence in distribution? Might not always be able to do this coupling
- Can we prove Donsker's directly from a (multidimensional) CLT, without needing this clever stopping time argument?

We'll say something about these today and next class (after spring break).

Definition 2.2. Let (S, d) be a complete separable metric space. $\mu_n \rightarrow \mu$ if $\int f d\mu_n \rightarrow \int f d\mu$ for all bounded and continuous functions $f : S \rightarrow \mathbb{R}$.

(**Added after lecture:**) given a probability measure μ on S and a measurable function $g : S \rightarrow S'$, the pushforward $g_{\#}\mu$ is the probability measure on S' given by

$$(g_{\#}\mu)(A) = \mu(g^{-1}(A))$$

for all measurable subsets $A \subseteq S'$. In other words, if $X \sim \mu$, then $g(X) \sim g_{\#}\mu$.

Theorem 2.3 (Continuous Mapping Theorem). *If $\mu_n \rightarrow \mu$ in distribution and $g : S \rightarrow S'$ continuous, then also $g_{\#}\mu_n \rightarrow g_{\#}\mu$ in distribution. This directly follows from the fact that if $\phi : S' \rightarrow \mathbb{R}$ is bounded continuous, then $\phi \circ g : S \rightarrow \mathbb{R}$ is as well.*

Theorem 2.4 (Portmanteau Theorem). *The following are equivalent:*

- (a) $\mu_n \rightarrow \mu$
- (b) $\int f d\mu_n \rightarrow \int f d\mu$ for bounded Lipschitz f .
- (c) $\forall C \subseteq S$ closed, $\limsup_{n \rightarrow \infty} \mu_n(C) \leq \mu(C)$.
- (d) $\forall U \subseteq S$ open, $\limsup_{n \rightarrow \infty} \mu_n(U) \geq \mu(U)$.
- (e) $\mu_n(A) \rightarrow \mu(A)$ if A is a measurable set with $\mu(\partial A) = 0$.

Proof. Some easy implicatures are (a) $\rightarrow$ (b) and (c) $\leftrightarrow$ (d). Another one is (c), (d) $\rightarrow$ (e). Let the closure be $\bar{A}$ and interior be A° . We see $\mu(\bar{A}) = \mu(A^\circ)$ and $\mu_n(\bar{A}) \geq \mu_n(A^\circ)$ as $\bar{A} \supseteq A \supseteq A^\circ$. Then we see $\limsup \mu_n(\bar{A}) \leq \mu(\bar{A}) = \mu(A^\circ) \leq \liminf \mu_n(A^\circ)$, so we easily conclude that everything is equal.

For (b) $\rightarrow$ (c), the idea is to approximate the indicator I_C from above by continuous functions (think about a hump with round corners). An explicit construction in a general metric space is

$$g_\epsilon(x) = \frac{d(x, (C^\epsilon)^c)}{d(x, (C^\epsilon)^c) + d(x, C)}$$

where $C^\epsilon = \{x, d(x, C) \leq \epsilon\}$ and $(\cdot)^c$ denotes complement. This is Lipschitz because both distances are Lipschitz in x and the denominator is always $\geq \epsilon$. By definition, we have $\int g_\epsilon d\mu_n \rightarrow \int g_\epsilon d\mu$ for all ϵ with $1 \ll 1/\epsilon \ll n$. For ϵ small, we have $\int g_\epsilon d\mu \rightarrow \int g d\mu$ by dominated convergence with $\int g_\epsilon d\mu_n \geq \int I_C d\mu_n$.

For (e) $\rightarrow$ (a), assume $f : S \rightarrow [0, 1]$ is continuous. Then

$$\int f d\mu = \int_0^1 \mu(\{x \in S : f(x) \geq y\}) dy$$

and

$$\int f d\mu_n = \int_0^1 \mu_n(\{x \in S : f(x) \geq y\}) dy.$$

Letting $A_y = \{x \in S : f(x) \geq y\}$, it is not hard to show that $\partial A_y \subseteq f^{-1}(y)$. This means $\mu(A_y) = 0$ except for countably many y . So by dominated convergence, the bottom integral converges to the top one (since the integrands are $[0, 1]$ -valued and countable sets have Lebesgue measure 0). $\square$